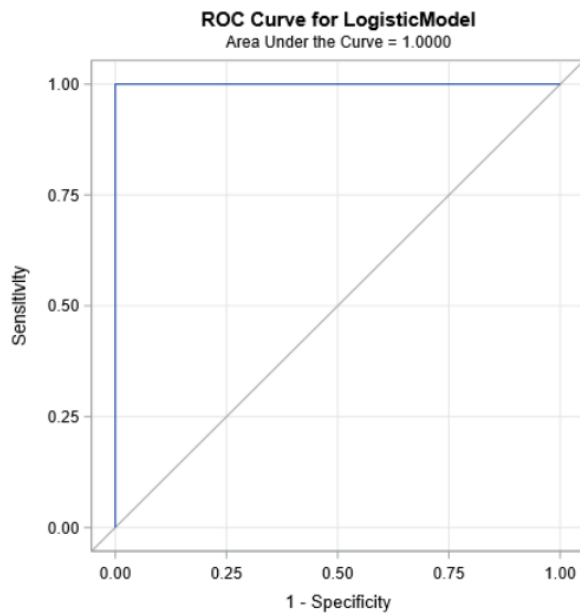


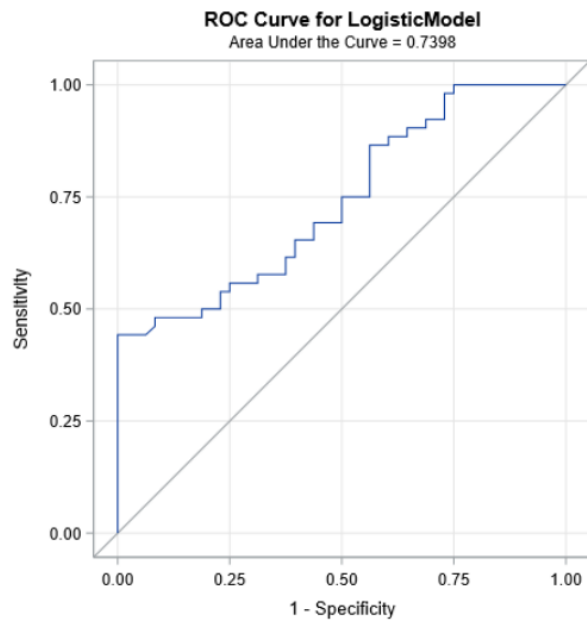
Homework 2
Stat 419
Alex Caramanico

1.
SAS Outputs

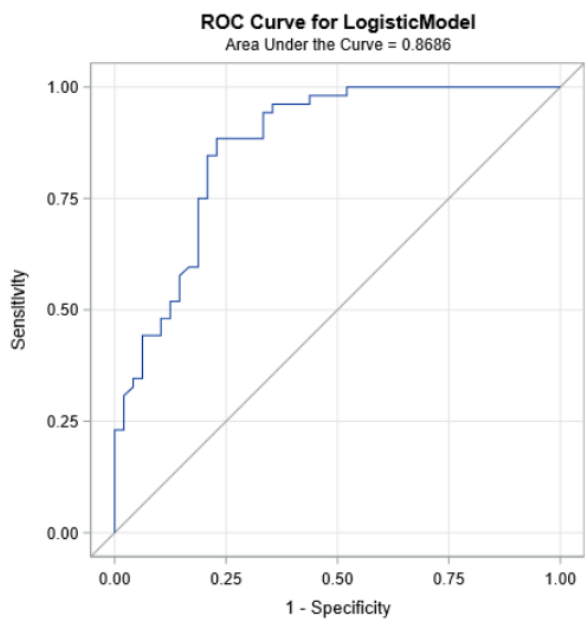
Test 1



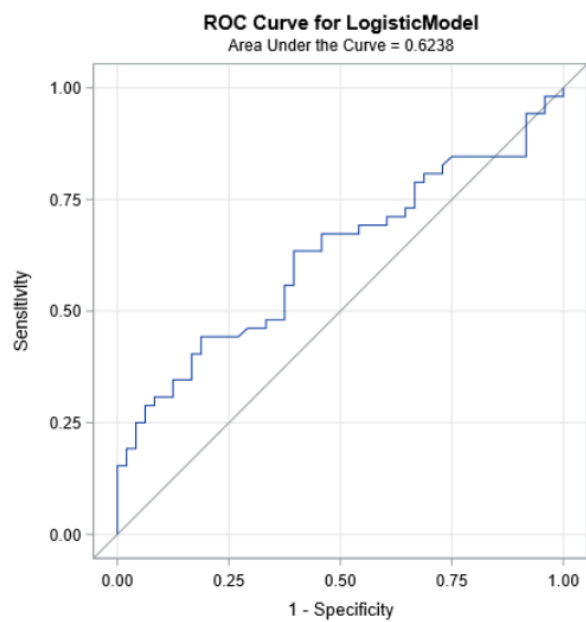
Test 2



Test 3



Test 4



Rankings

Rank	Test	Reasoning
1	1	Area under the curve of 1, in other words the tests is perfectly accurate when giving positive results and when giving negative results
2	3	Very high area under the curve, meaning the test accurately identifies the disease most of the time, with very few errors
3	2	Third highest area under the curve, correctly identifies disease usually but not as often as Tests 1 or 3
4	4	Lowest area under the curve, test has the most relative “false negatives” and “false positives”, i.e. it does not detect the disease well

Based on the results, it may be recommended to use Test 1 because it is the most accurate. That is, its sensitivity and specificity are always 1. This can be seen in the ROC plot generated from SAS created two straight-lines - one along the y-axis (where $1 - \text{specificity} = 0$), and one along $\text{sensitivity} = 1$. This is to say that the test always predicts those confirmed to HAVE the condition as positive, and always predicts those confirmed NOT to have the condition as negative. It can also be seen that the area under the curve (AUC) of Test 1 is 1, the maximum value it can be. While these results may not extend to the entire population (it probably is not reasonable to assume that observations from a sample, even a large one, will apply to each and every potential observation), the results seen in the sample can lead to a confident assertion that Test 1 would be the most useful.

2.

SAS Outputs for Frequency Tables

Table of diag1 by status			
diag1	status		Total
	0	1	
0	36	0	36
	36.00	0.00	36.00
	100.00	0.00	
	75.00	0.00	
1	12	52	64
	12.00	52.00	64.00
	18.75	81.25	
	25.00	100.00	
Total	48	52	100
	48.00	52.00	100.00

Table of diag2 by status			
diag2	status		Total
	0	1	
0	19	32	51
	19.00	32.00	51.00
	37.25	62.75	
	39.58	61.54	
1	29	20	49
	29.00	20.00	49.00
	59.18	40.82	
	60.42	38.46	
Total	48	52	100
	48.00	52.00	100.00

Table of diag3 by status			
diag3	status		Total
	0	1	
0	37	7	44
	37.00	7.00	44.00
	84.09	15.91	
	77.08	13.46	
1	11	45	56
	11.00	45.00	56.00
	19.64	80.36	
	22.92	86.54	
Total	48	52	100
	48.00	52.00	100.00

Table of diag4 by status			
diag4	status		Total
	0	1	
0	4	4	8
	4.00	4.00	8.00
	50.00	50.00	
	8.33	7.69	
1	44	48	92
	44.00	48.00	92.00
	47.83	52.17	
	91.67	92.31	
Total	48	52	100
	48.00	52.00	100.00

Test	1	2	3	4
Sensitivity	$\frac{52}{52+0} = 1$	$\frac{20}{20+32} = 0.385$	$\frac{45}{45+7} = 0.865$	$\frac{48}{48+4} = 0.923$
Specificity	$\frac{36}{36+12} = 0.75$	$\frac{19}{19+29} = 0.396$	$\frac{37}{37+11} = 0.771$	$\frac{4}{4+44} = 0.083$

Looking at these proportions, there may not be a clear answer as to which test is the best. Both Tests 1 and 3 appear to be particularly useful, as they are high in both categories. This means that the tests are effective at correctly identifying the disease (sensitivity) and correctly NOT identifying the disease (specificity). Tests 2 and 4 on the other hand are relatively weak in both aspects and much stronger in one aspect than the other, respectively. Test 2 makes many mistakes, but is fairly even in terms of its “false positive” and “false negative” rates - its sensitivity and specificity are very close to equal. Test 4 on the other hand is extremely good at identifying the disease, but not good at predicting against the disease. This can be seen by the fact it only predicted “negative” 8 times out of 100, which probably means its diagnosis criteria was extremely low, in turn meaning the vast majority of participants tested positive. For these reasons, it might be recommended to use either Test 1 or Test 3 to identify the disease among the general population, as they identify it with high accuracy without giving too many “false positives” or “false negatives”. Further, the first test makes no errors in detecting the disease when it is there, but does commit a few false positives. Since its sensitivity is higher than Test 3 without sacrificing a significant amount of specificity, it would probably be the first choice for disease diagnosis based on the information provided.

An interesting aspect to consider is using the test for different types of screening procedures. Based on the results, Test 1 is likely the most reliable in giving the most accurate result. That is, an accurate negative prediction or an accurate positive prediction. Test 3 is also solid in both of these areas, but commits slightly more “false negatives”. Test 4 may be useful in a situation where the researcher errs on the side of caution, and is willing to risk higher false positive rates so that false negative rates are limited. Test 2 is difficult to argue for, as it commits nearly as many false positive errors as it does false negative errors.

3.

Tests with large AUCs are generally good tests, in that they are more accurate in detecting a condition when it is present, and not detecting a condition when it is not present. If the purpose of the test is to accurately detect a condition, then it is difficult to argue against the efficacy of said test if it accomplishes said goal. However, the sensitivity and specificity of the test are always dependent upon the testing criteria. That is, how “severe” or clear a signal has to be in order for the test to predict a positive result for any given patient. *High* testing criteria, or one that requires clearer signals of the condition, leads to higher specificity because a *lower* proportion of the sample can test positive. On the other end, *low* testing criteria, or one which only requires minimal signals of the condition, leads to higher sensitivity because a *higher* proportion of the sample can test positive. This is to say that the efficacy of a clinical test can largely depend on where the testing criteria is set. Determining the “best” diagnostic value by which to create an ROC curve is challenging and often complex, because it is not ideal for everyone to test positive, nor everyone to test negative because of extreme criteria. Rather, it is necessary to find a sufficient middle ground between the two, which essentially turns the issue into a multi-variable optimization problem (with perhaps 3 or more variables).

Other crucial factors to consider are the invasiveness, cost, and administration techniques of the test. A certain test may be far more accurate than another, but in these cases it is common for said test to be invasive or uncomfortable or generally displeasing for the participant. In these cases, people would be less willing to take it, decreasing the sample size and increasing dropout rates, thus decreasing test precision. In addition, if it is extremely expensive to carry out the test, it may not be economically sustainable. While a test may be perfectly accurate, if it costs thousands to run once and/or requires a significant amount of time, then the potential sample size (and thus application to the entire population potentially suffering from the given condition) is once again inhibited.

SAS Code:

Data imported and called from separate program within SAS

data stat419hw2;

input status test1 diag1 test2 diag2 test3 diag3 test4 diag4;

Problem 1

proc logistic data=stat419hw2 plots(only)=roc;

LogisticModel: model status(event='1')=test1; # "test1" replaced with tests 2-4 for each curve

output out=LogiOut predicted=LogiPred;

run;

Problem 2

proc sort data=stat419hw2;

by descending diag1 descending status; # "diag1" replaced with diag 2-4 for each table

run;

proc freq;

tables diag1*status; # "diag1" replaced with diag 2-4 for each table

run;